Scalability & Future Plan

Date	15 JULY 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine Using Machine Learning
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	**Static File-based Data Storage**	Storing the agricultural dataset as CSV files on the local system limits scalability, real-time updates, and efficient data management.	High
2	**Limited Recommendation Explanation**	The system recommends a crop but does not explain why it was selected or suggest alternative crops based on the given soil and weather conditions.	High
3	**Synchronous In-Memory Inference**	The Flask application processes one prediction request at a time, which may reduce performance when many users access the application simultaneously	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Flask application running on a local development server with limited support for multiple users.	Migrate to a containerized Deploy the application using Docker and cloud platforms such as AWS, Azure, or Render with auto-scaling support.
2	Data Storage	Crop recommendation dataset stored as CSV files and accessed using Pandas.	Store agricultural data in a relational database such as PostgreSQL or MySQL for faster access and easier management.
3	Performance	Each crop prediction is processed individually using the trained ml model.	Use caching, asynchronous task processing, and optimized model serving to improve prediction speed.
4	Security	The application currently has basic security and does not include user authentication or encrypted communication.	Implement secure login, HTTPS communication, role-based access, and encrypted data storage.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	**Smart Crop Recommendation & Validation**	2026 (1-2 Months)	Provide explanations for crop recommendations and validate user inputs before prediction
Phase 2	**Database Integration & REST APIs**	2026 (3-4 Months)	Integrate agricultural databases, weather APIs, and soil information services for real-time recommendations.
Phase 3	**Automated Drift & Retraining Pipeline**	2026(5-6 Months)	Monitor changes in agricultural data, weather conditions, and crop patterns to automatically retrain the machine learning model and improve prediction accuracy..